HTML SVG

A Practical, From-Zero Learning Guide

Part 4 — Colors & Advanced Painting

SVG is not limited to flat colors. It supports gradients, patterns, transparency, color inheritance and advanced stroke effects. These features turn basic geometry into polished illustrations and UI graphics.

Goal: Learn how SVG paints shapes, then build reusable visual styles using gradients, patterns, opacity and CSS.

1. How SVG Paints a Shape

The two main painting properties are **fill** for the interior and **stroke** for the outline.

```
<circle cx="100" cy="100" r="60"
        fill="orange"
        stroke="black"
        stroke-width="6"/>
```

2. CSS Colors

```
<rect x="20" y="20" width="80" height="60" fill="red"/>
<rect x="120" y="20" width="80" height="60" fill="#2563eb"/>
<rect x="220" y="20" width="80" height="60" fill="rgb(34 197 94)"/>
<rect x="320" y="20" width="80" height="60" fill="hsl(45 90% 55%)"/>
```

• Named colors

• Hex colors

• RGB / RGBA

• HSL / HSLA

• CSS custom properties

3. currentColor

currentColor uses the CSS **color** value and is excellent for reusable icons.

```
<style>
.icon { color: #2563eb; }
</style>

<svg class="icon" viewBox="0 0 100 100">
  <circle cx="50" cy="50" r="35"
          fill="none"
          stroke="currentColor"
          stroke-width="8"/>
</svg>
```

4. CSS Variables

```
:root {
  --brand: #7c3aed;
  --accent: #f59e0b;
}
.icon {
  fill: var(--brand);
  stroke: var(--accent);
}
```

Variables are especially useful when your website has themes or a design system.

5. fill: none

```
<circle cx="100" cy="100" r="60"
        fill="none"
        stroke="black"
        stroke-width="8"/>
```

6. Stroke Line Caps

```
<line x1="30" y1="50" x2="270" y2="50"
      stroke="black" stroke-width="20"
      stroke-linecap="butt"/>
```

```
<line x1="30" y1="100" x2="270" y2="100"
      stroke="black" stroke-width="20"
      stroke-linecap="round"/>
<line x1="30" y1="150" x2="270" y2="150"
      stroke="black" stroke-width="20"
      stroke-linecap="square"/>
```

The three common values are **butt**, **round** and **square**.

7. Stroke Line Joins

```
<polyline points="40,160 100,40 160,160 220,40 280,160"
          fill="none"
          stroke="black"
          stroke-width="20"
          stroke-linejoin="round"/>
```

• miter — sharp corner

• round — rounded corner

• bevel — cut-off corner

8. Dashed Strokes

```
<line x1="20" y1="100" x2="280" y2="100"
      stroke="black"
      stroke-width="8"
      stroke-dasharray="20 10"/>
```

stroke-dasharray alternates dash and gap lengths.

9. stroke-dashoffset

```
<path d="M20 100 H280"
      fill="none"
      stroke="black"
      stroke-width="8"
      stroke-dasharray="30 10"
      stroke-dashoffset="15"/>
```

It shifts the dash pattern along the path and becomes very useful for path-drawing animations.

10. Opacity

```
<circle cx="80" cy="100" r="60"
        fill="red" opacity="0.5"/>
<circle cx="150" cy="100" r="60"
        fill="blue" fill-opacity="0.5"/>
```

opacity affects the whole element; **fill-opacity** affects only the fill; **stroke-opacity** affects only the stroke.

11. Linear Gradients

```
<svg viewBox="0 0 300 150">
  <defs>
    <linearGradient id="sky" x1="0" y1="0" x2="1" y2="1">
      <stop offset="0%" stop-color="#38bdf8"/>
      <stop offset="100%" stop-color="#1d4ed8"/>
    </linearGradient>
  </defs>
  <rect x="20" y="20" width="260" height="110"
        rx="20" fill="url(#sky)"/>
</svg>
```

A gradient is defined in **<defs>** and applied with **url(#id)**.

12. Gradient Stops

```
<linearGradient id="sunset">
  <stop offset="0%" stop-color="#f97316"/>
  <stop offset="50%" stop-color="#ec4899"/>
  <stop offset="100%" stop-color="#4f46e5"/>
</linearGradient>
```

Each **stop** has an offset and a color. You can use many stops.

13. Gradient Direction

```
<linearGradient id="horizontal"
                  x1="0%" y1="0%"
                  x2="100%" y2="0%">
  <stop offset="0%" stop-color="red"/>
  <stop offset="100%" stop-color="blue"/>
</linearGradient>
```

This gradient runs from left to right.

14. gradientUnits

```
<linearGradient id="g"
                  x1="0" y1="0"
                  x2="1" y2="1"
                  gradientUnits="objectBoundingBox">
  <stop offset="0%" stop-color="white"/>
  <stop offset="100%" stop-color="black"/>
</linearGradient>
```

objectBoundingBox uses the painted object's bounds. **userSpaceOnUse** uses the SVG coordinate system.

15. Radial Gradients

```
<radialGradient id="glow"
                  cx="50%" cy="50%" r="50%">
  <stop offset="0%" stop-color="white"/>
  <stop offset="60%" stop-color="#38bdf8"/>
  <stop offset="100%" stop-color="#1e3a8a"/>
</radialGradient>

<circle cx="150" cy="100" r="80"
        fill="url(#glow)"/>
```

Radial gradients are useful for glows, spheres and lighting effects.

16. Gradient Transform

```
<linearGradient id="rotated"
                  gradientTransform="rotate(45)">
  <stop offset="0%" stop-color="red"/>
  <stop offset="100%" stop-color="blue"/>
</linearGradient>
```

gradientTransform transforms the gradient itself.

17. Patterns

```
<defs>
  <pattern id="dots" width="30" height="30"
           patternUnits="userSpaceOnUse">
    <circle cx="8" cy="8" r="4" fill="#2563eb"/>
  </pattern>
</defs>
```

```
<rect x="10" y="10" width="280" height="160"
      fill="url(#dots)"/>
```

Patterns repeat SVG shapes across the painted area and are useful for dots, grids, stripes and textures.

18. Stripe Patterns

```
<pattern id="stripes"
         width="20" height="20"
         patternUnits="userSpaceOnUse"
         patternTransform="rotate(30)">
  <rect width="10" height="20" fill="#f97316"/>
</pattern>
```

A pattern can itself be transformed.

19. CSS Styling and Hover

```
<style>
.primary {
  fill: #7c3aed;
  stroke: #4c1d95;
  stroke-width: 4;
}
.primary:hover {
  fill: #8b5cf6;
}
</style>

<svg viewBox="0 0 200 150">
  <rect class="primary" x="40" y="30"
        width="120" height="90" rx="15"/>
</svg>
```

Inline SVG participates in the HTML/CSS environment, making interactive graphics practical.

20. Paint Order

```
<circle cx="90" cy="90" r="60" fill="red"/>
<circle cx="130" cy="90" r="60" fill="blue"/>
```

Later graphical elements are generally painted above earlier ones.

21. display and visibility

```
<circle cx="50" cy="50" r="30"
        fill="red" display="none"/>
<circle cx="130" cy="50" r="30"
        fill="blue" visibility="hidden"/>
```

display:none removes rendering. **visibility:hidden** keeps the geometry but hides the element.

22. Color Inheritance

```
<g color="#2563eb">
  <path d="M20 100 H100"
        stroke="currentColor" stroke-width="8"/>
  <circle cx="150" cy="100" r="30"
          fill="currentColor"/>
</g>
```

This is a powerful pattern for SVG icons whose color should follow their parent.

23. One Gradient, Many Shapes

```
<defs>
  <linearGradient id="brandGradient"
                  x1="0%" y1="0%"
                  x2="100%" y2="100%">
    <stop offset="0%" stop-color="#06b6d4"/>
    <stop offset="100%" stop-color="#7c3aed"/>
  </linearGradient>
</defs>

<circle cx="100" cy="90" r="60" fill="url(#brandGradient)"/>
<rect x="200" y="30" width="120" height="120"
      rx="20" fill="url(#brandGradient)"/>
<path d="M370 140 Q420 20 470 140 Z"
      fill="url(#brandGradient)"/>
```

A single definition can be reused by many elements.

24. Mini Project — Gradient Button

```
<svg viewBox="0 0 500 160">
  <defs>
    <linearGradient id="buttonGradient"
                    x1="0%" y1="0%" x2="100%" y2="0%">
      <stop offset="0%" stop-color="#7c3aed"/>
      <stop offset="100%" stop-color="#ec4899"/>
    </linearGradient>
  </defs>

  <rect x="50" y="35" width="400" height="90"
        rx="45" fill="url(#buttonGradient)"/>

  <circle cx="120" cy="80" r="25"
          fill="white" opacity="0.2"/>
</svg>
```

25. Mini Project — Glowing Sun

```
<svg viewBox="0 0 300 300">
  <defs>
    <radialGradient id="sun">
      <stop offset="0%" stop-color="#fff7ed"/>
      <stop offset="45%" stop-color="#fbbf24"/>
      <stop offset="100%" stop-color="#f97316"/>
    </radialGradient>
  </defs>

  <circle cx="150" cy="150" r="65" fill="url(#sun)"/>

  <g stroke="#f59e0b" stroke-width="8"
     stroke-linecap="round">
    <line x1="150" y1="25" x2="150" y2="55"/>
    <line x1="150" y1="245" x2="150" y2="275"/>
    <line x1="25" y1="150" x2="55" y2="150"/>
    <line x1="245" y1="150" x2="275" y2="150"/>
  </g>
</svg>
```

26. Exercises

• Create a rectangle with a left-to-right gradient.

• Create a radial gradient that looks like a glowing light.

• Create a dotted pattern and use it as a background.

• Create diagonal stripes with patternTransform.

• Build an icon using currentColor.

• Create three shapes that share one gradient.

• Create a sunset using a multi-stop gradient.

• Create a button-like SVG using gradients and rounded geometry.

• Experiment with stroke-dasharray and stroke-dashoffset.

27. Quick Reference

Property / Element	Purpose
fill	Interior paint
stroke	Outline paint
fill-opacity	Fill transparency
stroke-opacity	Stroke transparency
opacity	Whole-element transparency
stroke-linecap	Shape of open stroke ends
stroke-linejoin	Shape of stroke corners
stroke-dasharray	Dash pattern
stroke-dashoffset	Shift dash pattern
<linearGradient>	Linear color transition
<radialGradient>	Radial color transition
<stop>	Gradient stop
<pattern>	Repeating paint pattern
currentColor	Uses CSS color
url(#id)	References a defined paint resource

28. What's Next?

Part 5 will cover **SVG Text & Typography**: <text>, <tspan>, positioning, alignment, fonts, baselines, letter spacing, text paths and typography projects.